

Stat 150 – Lecture 04

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1 Main idea

Poisson r.v. $X \sim \text{Poisson}(\lambda)$, with probability generating function $G_X(s)$ for $s \in [0, 1]$:

$$G_X(s) = \sum_{i=0}^{\infty} \frac{e^{-\lambda} \lambda^i}{i!} s^i = e^{-\lambda} \sum_{i=0}^{\infty} \frac{(\lambda s)^i}{i!} = e^{-\lambda} e^{\lambda s} = e^{\lambda(s-1)} = e^{-\lambda(1-s)}$$

2 Galton–Watson branching process (Pinsky & Karlin §3.8.1)

Let X_n be the number of individuals of a population.

We often assume $X_0 = 1$ (n is generation n) Given $X_n = i$ and labelling by $n_1, n_2, n_3, \dots, n_i$ Suppose each of these individuals gives birth to $\zeta_{n1}, \zeta_{n2}, \dots, \zeta_{ni}$ Then,

$$X_{n+1} = \sum_{j=1}^{X_n} \zeta_{nj}$$

Assume all ζ_{nl} , $n, l \in \mathbb{Z}^+$, are independent and have the same distribution.

Generating function. Let $G(s) = \mathbb{E}[s^\zeta]$ be the offspring PGF ($s \in [0, 1]$), with mean $\mu = G'(1)$ and variance σ^2 , and let $G_n(s) = \mathbb{E}[s^{X_n}]$ (so $G_0(s) = s$, since $X_0 = 1$). A sum of m i.i.d. copies of ζ has PGF $G(s)^m$, so by the tower property,

$$G_{n+1}(s) = \mathbb{E}[\mathbb{E}[s^{X_{n+1}} \mid X_n]] = \mathbb{E}[G(s)^{X_n}] = G_n(G(s)).$$

Each generation just composes G one more time: $G_n = G \circ \dots \circ G$ (n times). Differentiating at $s = 1$ (where $G(1) = 1$, $G'(1) = \mu$) gives $\mathbb{E}[X_n] = \mu^n$.

Two identities. Everything above runs on two facts.

1. *Tower property:* $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$ — condition on Y , then average over Y , and you are back at the plain average:

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \sum_y \mathbb{E}[X \mid Y=y] \mathbb{P}(Y=y) = \sum_x x \sum_y \mathbb{P}(X=x, Y=y) = \mathbb{E}[X].$$

2. *Eve's law*: $\text{Var}(Z) = \mathbb{E}[\text{Var}(Z \mid Y)] + \text{Var}(\mathbb{E}[Z \mid Y])$. Condition the identity $\mathbb{E}[Z^2] = \text{Var}(Z) + (\mathbb{E}Z)^2$ on Y :

$$\mathbb{E}[Z^2 \mid Y] = \text{Var}(Z \mid Y) + (\mathbb{E}[Z \mid Y])^2,$$

average over Y , and subtract $(\mathbb{E}Z)^2 = (\mathbb{E}[\mathbb{E}[Z \mid Y]])^2$; the last two terms are then exactly the variance of the random variable $\mathbb{E}[Z \mid Y]$:

$$\text{Var}(Z) = \mathbb{E}[\text{Var}(Z \mid Y)] + \text{Var}(\mathbb{E}[Z \mid Y]).$$

Variance of X_{n+1} . Given X_n , the next generation is a sum of X_n i.i.d. copies of ζ , so linearity and independence give

$$\mathbb{E}[X_{n+1} \mid X_n] = \mu X_n, \quad \text{Var}(X_{n+1} \mid X_n) = \sigma^2 X_n,$$

and Eve's law turns these into a recursion:

$$\text{Var}(X_{n+1}) = \mathbb{E}[\sigma^2 X_n] + \text{Var}(\mu X_n) = \sigma^2 \mathbb{E}[X_n] + \mu^2 \text{Var}(X_n).$$

When $\mu = 1$ this gives $\text{Var}(X_n) = n\sigma^2$; in general $\text{Var}(X_n) = \sigma^2 \mu^{n-1} \frac{\mu^n - 1}{\mu - 1}$ for $\mu \neq 1$.

3 Reference: notation and facts to remember

Notation.

- $X \sim \text{Poisson}(\lambda), \text{Bin}(n, p), \text{Bern}(p), \text{Geom}(p), \text{Exp}(\lambda)$: “ X has this distribution.”
- i.i.d.: independent and identically distributed.
- $X \perp\!\!\!\perp Y$: independent. $\mathbb{P}(A \mid B) = \mathbb{P}(A, B)/\mathbb{P}(B)$, where “,” means “and”.
- $\mathbf{1}_A$ (indicator): 1 if A happens, 0 otherwise; note $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$.
- $G_X(s) = \mathbb{E}[s^X]$: probability generating function, $s \in [0, 1]$.
- $\mathbb{E}[X \mid Y=y]$ is a *number* (the average of X over outcomes with $Y = y$); $\mathbb{E}[X \mid Y]$ is a *random variable*, a function of Y .
- This unit: X_n = size of generation n , ζ = offspring count, $\mu = \mathbb{E}[\zeta]$, $\sigma^2 = \text{Var}(\zeta)$, G = offspring PGF.

Expectation.

- Linearity — no independence needed: $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$; sums always split, even with a random number of (independent) terms: $\mathbb{E}[X_1 + \dots + X_N] = \mathbb{E}[N] \mathbb{E}[X]$.
- $\mathbb{E}[c] = c$; $X \geq 0 \implies \mathbb{E}[X] \geq 0$.
- $X \perp\!\!\!\perp Y \implies \mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y]$ (the converse is false).
- Tower property: $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$.

Variance and covariance.

- $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2 = \mathbb{E}[(X - \mathbb{E}X)^2] \geq 0$.

- $\text{Var}(c) = 0$; $\text{Var}(aX) = a^2 \text{Var}(X)$ — note the square.
- $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$, symmetric, bilinear, $\text{Cov}(X, X) = \text{Var}(X)$.
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$.
- Independent $\implies \text{Cov} = 0 \implies$ variances add; n i.i.d. terms: $\text{Var}(X_1 + \dots + X_n) = n\sigma^2$.
- Eve's law: $\text{Var}(Z) = \mathbb{E}[\text{Var}(Z \mid Y)] + \text{Var}(\mathbb{E}[Z \mid Y])$.

Generating function facts.

- $G(1) = 1$; $G(0) = \mathbb{P}(X = 0)$; $G'(1) = \mathbb{E}[X]$; $\text{Var}(X) = G''(1) + G'(1) - (G'(1))^2$.
- Independent sums multiply: $G_{X+Y} = G_X \cdot G_Y$; n i.i.d. copies give $G(s)^n$.
- Random number of terms: $N \perp\!\!\!\perp$ the X_i gives $G_{X_1+\dots+X_N}(s) = G_N(G_X(s))$ — this is the Galton–Watson recursion $G_{n+1}(s) = G_n(G(s))$.
- Memorize: Bern(p): $1 - p + ps$; Bin(n, p): $(1 - p + ps)^n$; Geom(p): $\frac{ps}{1 - (1-p)s}$; Poisson(λ): $e^{\lambda(s-1)}$.

4 Claims, theorems, and proof sketches

Claim. State the claim in one precise sentence.

Proof sketch. Record the key idea, the first useful equation, and the step that still needs checking.

5 Examples and calculations

Example 5.1 (Lecture example). Write the setup, the calculation, and the conclusion.

6 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?
- TODO: